



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

EDMS 01-200-1

المواصفات الفنية لموزع جهد متوسط ٢٤ ك. ف
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**TECHNICAL SPECIFICATIONS
FOR 24 KV SWITCH BOARD**

Issue: MAY-2016 / Rev- 1

❖ توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

I- Objective:

The Objective of this specification is to achieve the minimum requirements for the following :

Indoor medium voltage (24 K.V) A.C. metal clad switch board , 3 phase , (single / double) bus bar in accordance with the latest international standard IEC 62271-200, IEC 62271-100 unless otherwise specified .

It Shall be made of sheet steel and equipped with vacuum or SF6 sealed / rechargeable circuit breaker, copper bus bars and all necessary connections.

The board consists of (X) panels as follows:-

- (X1) as incoming feeders panels.
- (X2) as outgoing feeders panels.
- Bus-coupler section (2 panels) coupler & riser.

General data :

Rated voltage.	: 24 K.V
Insulation level at power frequency.	: 50 K.V.
Basic impulse level (BIL.)	: 125 K.V .
Frequency.	: 50 HZ.
Colour	: Preferred (light grey)
Ambient temperature	: 40°C.
Average Humidity level	: 90%



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

Auxiliary voltage.	: 110 V.D.C $\pm$ 20%
Degree of protection	: IP 4X
Copper purity	: not less than 99.9 %

Circuit Breakers :-

A- The 24 K.V. circuit breakers shall be vacuum or SF6 type.

B- The necessary closing , tripping and blocking coils shall be (110 V DC). With an electric spring charge motor drive (110 V DC) or magnetic actuator mechanism consist of three coils (one per pole) and all phases connected to each other using synchronizing shaft.

The circuit breakers shall be mechanically and electrically trip free and with provision for manual operation.

C- Necessary auxiliary contacts for controlling, signaling& free contacts (4NO-4NC) at least in low voltage compartment and all accessories needed for operation shall be provided.

D- The failure of C.B in switching off during any fault must be indicated through programmed LED from the relay.

E- All circuit breakers shall be fitted with an operation counter.

F- If :
- The C.B. is rechargeable type the SF6 pressure indication should be through two stages (Low – insufficient).
- The C.B. is sealed type one flag for SF6 insufficient is enough.
(for low gas pressure should be alarm and insufficient gas pressure should be blocked)

G- The minimum rupturing capacity of all the 24 K.V. circuit breakers should be symmetrical



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

short circuit current of 25 K.A. for 3 Sec at least.

H- The normal current ratings for C.B's are as follows:

Incoming C.B's : 1250 A.

Outgoing C.B's : 1250 A.

Bus Coupler C.B: 1250 A

I- All main fixed and moving contacts shall be silver plated according to IEC.

Bus Bars:

A- Main Bus Bars :

1. It shall be insulated B.B system and made of best quality, high conductivity copper with C.S.A. of $1 \times 120 \times 10$ mm or $2 \times 80 \times 10$ mm (for all panels) and the connectors (Droppers) shall not be less than 800 mm^2 .

The maximum allowable temperature rise shall be 25°C above the ambient temperature of 40°C .

2. The bus bar when passing from a cubicle to another shall be through an insulating bushing.
3. The bus-bar compartments shall be placed at the upper part of the switchboard and fitted with using dielectric supports.
4. All main B.Bs and Droppers shall be insulated by heat shrinkable tubes (Red Anti-Track) and the connection points with suitable insulated covers.
5. The entire bus and structure shall be constructed to withstand the short circuit effects due



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

to the rated short circuit current of 25 K.A for 3 sec at least.

- The incoming feeder shall be equipped with three surge arrestors (gap less metal oxide) 10 K.A, 24 KV max. continuous operating voltage $20 \text{ KV} \pm 5\%$ (leakage path 3 cm/kv at least) in case of vacuum C.B and installed in cable compartment.
- Main B.B shall be extendable and securely mechanically bonded to each unit.

B-Earth bus-bar:

An earth copper bar at the bottom of the switchboard extending the entire length of each distribution panel shall be provided with a cross section area to withstand the rated short circuit current of the system .The bus shall be extendable and securely mechanically bonded to each unit.

Instrument Transformers:

The following primary currents can be changed due to the requirements of electricity distribution company:

- In Incoming cubicles C.T burden shall be 10 VA class 0.5 for measuring and 15 VA class 5P10 for protection relays (The burden is achieved for both two ratios 300-600/5/5A).
- In coupler cubicle C.T burden shall be 10 VA class 0.5 for measuring and 15 VA class 5P10 for protection relay (The burden is achieved for both two ratios 600-1200/5/5A).
- In Outgoing cubicles C.T burden shall be 10 VA class 0.5 for measuring and 15 VA class 5P10 for protection relay (The burden is achieved for both two ratios 200-400/5/5A).
- The winding of the instrument transformers shall be made of copper.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

5. Current transformers for each cubicle shall be single phase dry type and capable of withstanding the rated short circuit current of 25 K.A for 3 sec at least. at rated voltage. The current transformers shall be interchangeable from lower ratio to higher ratio from the primary side or the secondary side of the current transformer.
6. Potential transformers shall be protected by suitable H.R.C fuses on the Primary side and they shall be single phase (withdrawable type as option) with the following ratios:
22000 / 110 / 110 V

$\sqrt{3}$ $\sqrt{3}$ 3

100 VA, cl 0.5 for first core and cl 3P for second core -

(voltage factor 1.9 for 30 sec.&1.2 continuous) - for each incoming feeder .

*In case of no need for open delta connection, the supplier shall close it by a suitable resistance.

II- Technical specifications:

1- Incoming feeder panel

Each consists of :

The cubicle with three phase copper bus-bars

(C.S.A. of $1 \times 120 \times 10 \text{ mm}^2$ or $2 \times 80 \times 10 \text{ mm}^2$) as the following:

- Vacuum or SF6 circuit breaker 1250A, at least 25 K.A for 3 Sec, 24KV, provided with motor operated 110 V.D.C, or (magnetic actuator) , with operating counter at the front .
- The operating sequence of C.B shall be
(O-0.3 sec – CO- 3 min- CO) or (O-0.3 sec – CO- 15 sec- CO).



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Closing time ≤ 60 ms.
- Opening time ≤ 50 ms.
- Low voltage plugs must be made of non-hygroscopic and non-flammable materials.
- Fixed isolating contacts.
- Three Single Phase current transformers 300-600/5/5 A (10 VA cl 0.5 for measuring & 15 VA cl 5 p 10 for Protection). The name plate shall be in the front, according to IEC.
- Three single phase (withdrawable type as option) potential transformers:
22000 / 110 / 110 V
 $\sqrt{3}$ $\sqrt{3}$ 3
100 VA, cl 0.5 for first core and cl 3P for second core -
(voltage factor 1.9 for 30 sec.&1.2 continuous) for each incoming feeder .
- In case of no need for open delta connection, the supplier shall close it by a suitable resistance.
- Three suitable H.R.C fuses 24 K.V.
- Earthing switch shall be capable to withstand the rated short circuit current 25 K.A for 1 Sec at least installed on the end cable side of current transformer not on the bus bar side.
- Low voltage socket.
- Low voltage plug.
- Three ammeters digital programmable type with suitable scale, accuracy class 0.5 fed from power supply of 110 V.DC.
- One voltmeter with scale (0-24) KV, digital type with accuracy class 0.5 fed from power supply 110 VDC and selector switch 7 positions.
- One D.C. position indicator for the circuit breaker.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- One D.C. position indicator for earthing switch.
- One DC position indicator in case of withdrawable P.T at secondary side.
- One (3 phase) 4-wire (KWH, KVAR, KW and power factor) digital meter, accuracy class 0.5 for active and 1 for reactive. (As option) .
- One (3-phase 4-wire) digital multi-meter at each panel to measure (As option):
 - (KWH accuracy class 0.5 and KVARH accuracy class 1)
 - 3 Ph current (IR – IS – IT – IN).
 - 3 Ph voltage (line to line and line to neutral) .
 - Power factor , THD and maximum demand .
- Three indicating lamps (LED TYPE) (on-off-trip) - (red - green - yellow)
- Two push buttons (on- off) electrical.
- Lamps for voltage phases (LI- L2- L3) (LED TYPE) .
- Three surge arrestors (gap less metal oxide) 10 K.A, 24 KV max. continues operating voltage 20 KV $\pm$ 5% (leakage path 3 cm/kv at least) in case of vacuum C.B and installed in cable compartment.
- One 3 phase combined directional and Non directional over current and earth fault protection relay according to the attached specifications.
- Two suitable space heaters , their capacities shall be as required to maintain the compartment and unit internal temperature above the dew point, taking into consideration the specified environmental condition, one in cable compartment and the other in C.B compartment for each panel + Hygrostat with suitable MCB.
- One flag relay with a reset push button for each of the following (D.C loss. - A.C loss – any M.C.B trip – low gas – insufficient gas-protective relay trip) and alarm.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

2- Bus bar coupler panel:

Consists of:

The cubicle with three phase copper bus-bars (C.S.A. of $1 \times 120 \times 10 \text{ mm}^2$ or $2 \times 80 \times 10 \text{ mm}^2$) as the following:-

- Vacuum or SF6 circuit breaker 1250A, at least 25 K.A for 3 Sec, 24 K.V, provided with motor operated 110 V.D.C, or (magnetic actuator) , with operating counter at the front.
- The operating sequence of C.B shall be (O-0.3 sec – CO- 3 min- CO) or (O-0.3 sec – CO- 15 sec- CO).
- Closing time $\leq 60 \text{ ms}$.
- Opening time $\leq 50 \text{ ms}$.
- Low voltage socket
- Fixed isolating contact
- Low voltage plug
- Three ammeters digital programmable type with suitable scale, with accuracy class 0.5 fed from power supply of 110 V.DC.
- One DC position indicator for the circuit breaker.
- One flag relay with a reset push button and alarm for each of the following (D.C loss. - A.C loss – any M.C.B trip – low gas – insufficient gas-protective relay trip).
- Suitable space heaters , its capacity shall be as required to maintain the compartment and unit internal temperature above the dew point, taking into consideration the specified environmental condition + Hygrostat with suitable M.C.B.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- One horn fed from power supply 110 VDC to give alarm during the following cases:-
 - a. DC loss in any panel.
 - b. Protection relays trip.
 - c. Low gas pressure.
 - d. Insufficient gas pressure.
 - e. AC incoming feeder 22 K.V loss for any incoming panel
 - f. Any M.C.B trip
- Three indicating lamps (LED type) (on-off-trip)-(red-green-yellow) .
- Two push buttons (on-off)-electrical operation.
- One 3 -phase combined non-directional over current- short circuit & earth fault relay according to the attached specifications.
- Three single phase potential transformers withdrawable type (as option)

22000 / 110 / 110 V

$\sqrt{3}$ $\sqrt{3}$ 3

100 VA, cl 0.5 for first core and cl 3P for second core -

(voltage factor 1.9 for 30 sec.&1.2 continuous).

- One voltmeter with scale (0-24) KV, digital type with accuracy class 0.5 fed from power supply 110 VDC and selector switch 7 positions (as option) .

Bus riser Panel Consists of:

The cubicle with three phase copper bus-bars

(C.S.A. of $1 \times 120 \times 10 \text{ mm}^2$ or $2 \times 80 \times 10 \text{ mm}^2$) as the following:



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Three Current transformer 600-1200/5/5A (10 VA cl 0.5 for measuring & 15VA cl 5 p10 for protection) with name plate at the front, according to IEC.
- Three single phase potential transformers withdrawable type (as option).
 $\frac{22000}{\sqrt{3}} / \frac{110}{\sqrt{3}} / \frac{110}{3} V$
100 VA, cl 0.5 for first core and cl 3P for second core -
(Voltage factor 1.9 for 30 sec.&1.2 continuous) .
- One voltmeter with scale (0-24) KV, digital type with accuracy class 0.5 fed from power supply 110 VDC and selector switch 7 positions (as option) .

3- Outgoing feeder panels:

Each consists of :

The cubicle with three phase copper bus-bars

(C.S.A. of $1 \times 120 \times 10 \text{ mm}^2$ or $2 \times 80 \times 10 \text{ mm}^2$) as the following:

- Vacuum or SF6 circuit breaker 1250A, at least 25 K.A for 3 Sec, 24 K.V, provided with motor operated 110 V.D.C, or (magnetic actuator) , with operating counter at the front.
- The operating sequence of C.B shall be
 - (O-0.3 sec – CO- 3 min- CO) or (O-0.3 sec – CO- 15 sec- CO).
 - Closing time $\leq 60 \text{ ms}$.
 - Opening time $\leq 50 \text{ ms}$.
- Fixed isolating contacts.
- Three single phase current transformers 200-400/5/5 A (10 VA cl 0.5 for measuring &



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

15 VA cl 5 p10 for Protection) , The name plate shall be in the front, according to IEC.

- One earthing switch shall be capable to withstand the rated short circuit current 25 K.A for 1 Sec installed on the end cable side of current transformer not on the bus bar side.
- Low voltage socket
- Low voltage plug.
- Three ammeters digital programmable type with suitable scale ,accuracy class 0.5 fed from power supply of 110 V.DC.
- One (3 phase) 4-wire (KWH , KVAR,KW and power factor) digital meter, accuracy class 0.5 for active and 1 for reactive. (As option) .
- One (3-phase 4-wire) digital multi-meter at each panel to measure (option):
 - (KWH accuracy class 0.5 and KVARH accuracy class 1)
 - 3 Ph current (IR – IS – IT – IN).
 - 3 Ph voltage (line to line and line to neutral) .
 - Power factor , THD and maximum demand .
- One D.C position indicator for the circuit breaker.
- One D.C position indicator for the earthing switch.
- Three indicating lamps (on-off-trip) - (red- green -yellow) (LED type).
- Two push buttons (on- off) – electrical operation.
- One 3 -ph combined Non direction over current - short circuit & Earth fault Protection relay according to the attached specifications.
- Two suitable space heaters , its capacity shall be as required to maintain the compartment and unit internal temperature above the dew point, taking into consideration the specified environmental condition, one in cable compartment and the other in C.B compartment for each panel + Hygrostat with suitable MCB.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- One flag relay with reset push button and alarm for each of the following :
(D.C loss- any M.C.B trip- low gas - insufficient gas-protective relay trip).

III- Design Criteria :

1. The switchboard shall consist of metal-clad cubicles of air insulated switchgear withdrawable vacuum or SF6 C.B.
2. Each cubicle has to be built as a self-supporting structure from rolled steel sheets of 2.5 mm thickness before paint and the width of cubicle shall not be less than 90 cm. The cubicles are to be bolted together.
Each cubicle is divided into four sealed and isolated compartments:

The bus bar compartment, cable connection compartment, circuit-breaker compartment and a fully isolated low-voltage compartment.
3. When moving the circuit-breaker out from the operating position the primary fixed contacts coming from the other two compartments must be covered automatically by steel (insulating material as option) safety shutters, complete compartmentalization must be maintained in all operating condition of switchgear.
4. Each compartment should be internally arc proofed and should be provided with pressure rifle ,(Also, flipper circuit should be installed in each cell if distribution company required) .
5. Electrical interlock will be between any incoming feeder panel for each section and coupler.
6. Mechanical interlock must be introduced against the following:
 - A. Moving the truck with the circuit-breaker closed.
 - B. Switching the circuit-breaker in any other position except the "Locked" , "Operating" - or -



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- "Test" position.
- C. Moving the truck into the "OPERATING" position with the low voltage plug disengaged.
 - D. Engaging the earthing-switch-with the truck in-the "OPERATING" position and moving the truck into this position with the earthing switch engaged.
 - E. Opening of the C.B compartment door with the C.B closed.
 - F. Opening of the cable compartment door with the earthing switch opened(as option).
7. All metallic parts not carrying current must be connected to an earthing copper black-painted bus bar of suitable cross-sectional area. Earthing of circuit-breaker truck must be secured till it is completely moved outside its cubicle.
8. All equipments (push buttons– indicating lamps – sockets and plugs – terminal blocks for C.Ts & P.Ts and control ...etc) must be made of hygroscopic and non-flammable material.
9. a- All the circuit carrying current parts of the switch board component must be made of high grade copper, the windings of instrument transformer must be made of copper.
- b- Cable connections shall be put from the below rear or front of the cubicles arrangement must be enough for connection of three single phase aluminum cable of cross-sectional area of $1 \times 400 \text{ mm}^2$ or $3 \times 300 \text{ mm}^2$ to each incoming cubicle and the connection of three phase aluminum cable of different cross-sectional area up to $3 \times 300 \text{ mm}^2$ for each outgoing cubicles (or as distribution company requirement) .
- 10.a-Wiring of secondary circuits must be executed using 0.6/1 KV non-flammable PVC-insulated copper conductors of cross-sectional area not less than the following:-
- Voltage circuits 1.5 mm^2 with opening slides terminal block



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Current circuits 3 mm² with short circuits slides terminal block
- Main heaters circuits 6 mm²

b-The secondary circuits of circuit-breaker truck have to be connected by means of flexible copper wires into a flexible metallic conduit and by multi-pin plug and socket of suitable size.

c-The insulation of secondary circuit must withstand at least a test voltage 2 kV, 50 c/s for 1 minute.

11. All relays must be of D.C. version i.e timing relays, auxiliary relays and tripping relays must be 110 V.D.C operated.
12. All the contacts in the control, protection and signaling circuits must be properly selected to be capable to interrupt and switch on the currents flowing through them during their operation.
13. A. The voltage secondary circuits must be protected by suitable automatic M.C.B.
B. The universal motor shall be protected by 2 pole MCB.
14. The metal-clad structure shall be treated against rust, and provided with priming, intermediate and finishing coats of electrostatic painting powder. The external coat is generally light grey color.
15. The instruments shall be of flush type and mounted on hinged door at the front of each cubicle.
16. The switch board has to be designed to give maximum reliability and safety in service concerning operation, inspection and maintenance.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

17. The switchboard has to be designed to be safe against spread of fire, formation of internal excessive pressure (each compartment shall be provided with its own pressure relief device), contact of live parts, ingress of dust and splashing liquid, thus degree of protection has to be not less than IP4x.
18. Every incoming and outgoing feeder, cubicle shall be equipped with an earthing switch.
19. The construction of switchboard and the C.B's shall comply with these specifications and IEC 62271-200, IEC 62271-100.
20. The leakage path for fixing insulators of P.T's and C.T's and shall not less than 44 cm.
21. The permissible (guaranteed) number of switching (ON/OFF) cycles of C.B's mechanism shall be ≥ 10000 cycle.
22. The indicating lamps & push button must be heavy duty.
23. Minimum phase to phase and phase to earth air clearance as distribution company required.
24. Indicating lamps (LED) for voltage phases (L1, L2, L3) shall be fed from capacitive divider.
25. All measuring instrument shall be digital type with range up to 140 % of their nominal rating.
26. The meters shall be supplied with their software and cables needed for communicate it with any computers.
27. All current transformers shall be provided with a current continuous factor =1.2.

IV- Drawings And Catalogues:

The tenderer must deliver with the offer a complete copy of outline drawings, single line diagram and a complete catalogues (and a wiring diagram if requested) for each component containing the full technical data in order to allow complete study of his offer.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

V- Guarantee Tables:

The tenderer must fill in all the guarantee tables attached with these specifications accurately.

The offer which will not be accompanied with a complete copy of drawings, catalogues and filling guarantee tables will not be considered.

VI- Marking :

Marking of the cubicles and their components must be done according to IEC standards.

VII- Testing :

a. Type tests should be as per the IEC 62271.

b. Routine tests:

Testing (routine tests) of all components shall be carried out according to latest IEC standards by the manufacturer, a representative of distribution company will attend these tests before acceptance – (attached list of the routine tests to be carried out).

Attendance of customer representative during tests mentioned in IEC may be complete or partial as it will be agreed, upon in the contract.

N.B:

In case of delivering C.B's of magnetic actuator type, it shall be delivered two control module units for each switch board free price.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

Protection Relays

I- Incoming Cubicles:

3- phase combined directional and Non-directional digital protection relay

The required is a combined directional and non-directional (over current + earth fault) relay and timer in one unit connected to three single phase current transformers and three single phase voltage transformers of the protected object and suitable for resistance earthed neutral system fulfilling the following requirements and specifications:

1- GENERAL CHARACTERISTICS:-

1.1-The protection relay is based on a multiprocessor.

1.2-The modern technology shall be applied in both hardware and software solution.

1.3-The relay must be designed for protection against (O.C.& E.F & S.C) in parallel feeders from S/S to distribution board and include the following:

- The current unit shall be supplied from 3-single phase current transformers 300 / 600/5/5 A ($I_R - I_S - I_T - I_N$)
- The voltage unit shall be supplied from 3-single phase voltage transformers (22000 / $\sqrt{3}$ /110/ $\sqrt{3}$ / 110/3) volt . ($V_R - V_S - V_T - V_0$)
- Nominal current $I_n = 5A / 1 A$, frequency 50 Hz
- Nominal voltage V_n (110-100) volt , frequency 50 Hz
- Reset factors $\geq 95\%$



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Operating temp. from -5°C to $+55^{\circ}\text{C}$.
- Humidity > 90% non-condensation
- Auxiliary supply : 110 V.D.C. $\pm$ 20 %
- It shall be able to operate without intended delay.
- Operation indication, with hand reset and automatic reset for tripping order and include one LED for each fault type and monitoring.
- Output contacts (Tripping contacts) .
 - Rated voltage 110 VDC $\pm$ 20% with ripple content (3-4) %.
 - Continuous carrying capacity 5 A DC.
 - Making and carrying current capacity for 0.5 sec $\geq$ 25 A .
 - Make and carry for 3.0 sec $\geq$ 10 A .
 - Free contacts at least (1 N.O for O.C & 1 N.O for E.F and 1 N.C for signaling).
 - Two trip contacts and two configurable per relay
- Phase discontinuity protection for three phases
- Three-phase thermal protection
- The relay must has the inrush current blocking based on the second order harmonic percentage which can be used in all protection stages , without using multiplayer
- All parameters and events can be set or read by P.C and manually by key pad
- Smaller dimensions preferred
- Tender must supply with offer detailed instruction and catalogue



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- The relay should be compatible with SCADA system (option)
- The relay must include event recorder and shall be read by screen.
- Original software, firmware, plug connections to (PC and / or laptop) for all protection relays and test plugs should provide by the contractor.
- Continuous self-diagnosis of electronics and software
- User-selectable password protection for HMI
- Display of primary / secondary (current & voltage) values
- Housing degree of protection IP 52 according to IEC 60529
- The relay should be supported by standard protocol IEC 60870-5-103 **if needed**
- The relay shall have two setting tables at least.
- Output contacts can be programmed by keypad and PC.
- Disturbance recorder with the following features:
 - Nonvolatile memory for disturbance record all data shall be kept even for switch off D.C. power supply.
 - Four analogue channels for (I_a , I_b , I_c , I_o) – (V_a , V_b , V_c , V_o) .
 - Eight digital channels at least .
 - The relay shall keep minimum of 5 records which can be stored.
 - Recording time capacity ≥ 15 sec.
 - Recording time up to three seconds duration for each storage fault.
 - Pre event and post event recording and memory time adjustable.
 - Oscillography recorder : voltage and current waves must be recorded so that can be analyzed oscillography through PC for all abnormal.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Fault recording from pre fault up to fault clearing stage with all electrical parameters.

1.4- Standard tests :

a. Insulation tests :

- Dielectric tests :
according to the (IEC 60255/5 & 60225/5 or equivalent) tests voltage 2 KV , 50 HZ, 60 sec.
- Impulse voltage test:
according to the (IEC 60255/5 & 60225/5 or equivalent) tests voltage 5KV, unipolar impulses, waveform 1.2/50 μ s
- Insulation resistance measurements:
according to the IEC60255/5

b. High frequency interference withstands:

According to class III IEC 60225/5

c. Over load

- Current circuit temporary > 60 In for 1.0 sec
- Polarization temporary > 1.3 Ur max (Ur = 3 Uo)

d. Mechanical tests

- Vibration test (sinusoidal) according to the IEC 62271-103.
- Shock and bump test according to the IEC 62271-103.

1. 5 The relay must be include the following protection:



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- $I >$ Three-phase low-set over current stage with definite-time and inverse definite minimum time characteristic (IDMT) (as option)
- $I \gg$ Three-phase high-set over current stage with the instantaneous or definite time characteristic.
- $I_o >$ low-set non-directional earth –fault stage with definite-time or inverse definite minimum time characteristic (IDMT).
- $I_o \gg$ High-set non-directional earth-fault stage with instantaneous or definite – time characteristic.

2-Parameterization:

The parameter can be set either locally over HMI or externally via the serial communication using relay software compatible with all new version of windows.

3-Nonvolatile memory:

The protection relay should be equipped with a nonvolatile memory for preserving important data during auxiliary supply breaks. The memory does not need batteries, and a lifelong service is guaranteed and Following data is stored:

- Date and Time
 - Setting values
 - Disturbance records
 - Record last five fault data (with all fault parameters).
 - At least, Last 50 events recorded
- Recorded values are stored from start, trip and other important external input .



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- At least, the last 50 events that occurred before the auxiliary power supply breaks can be viewed in the HMI event buffer. Time and date for the events are also restored.
- Disturbance recorder with the following features:
 - Nonvolatile memory for disturbance record all data shall be kept even for switch off D.C. power supply.
 - Four analogue channels for (Ia , Ib , Ic , Io) – (Va , Vb , Vc , Vo) .
 - Eight digital channels at least .
 - The relay shall keep minimum of 5 records which can be stored.
 - Recording time capacity ≥ 15 sec.
 - Recording time up to three seconds duration for each storage fault.
 - Pre event and post event recording and memory time adjustable.
 - Oscillography recorder : voltage and current waves must be recorded so that can be analyzed oscillography through PC for all abnormal.
 - Fault recording from pre fault up to fault clearing stage with all electrical parameters.

4-The real-time clock :

The relay must be supported by a real-time clock (RTC) which used for time stamping of events. It is also running during auxiliary power breaks.

When the supply is re-established, the relay sets the right time and new events are stamped accordingly.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

5-Setting values and ranges :

Nominal voltage (U_n): 100-110 v , F_n : 50 HZ and I_n : 5 A / 1A.

5.1. Three-phase directional over current protection :

- Setting current range : $(0.05 - 4) I_n$, with steps $\leq 0.05 I_n$
- Start voltage: $(2\%) U_n$
- Operate time in DT mode : $(0.05 : 3)$ sec with step ≤ 0.01 sec
- Basic angle ϕ ($0 : 359^\circ$ with step $\leq 5^\circ$) or $(-90^\circ : +90^\circ)$ with Operation direction (Forward & Reverse)
- Min. voltage of polarizing quantity for operation (sensitivity) shall be $\leq 2\% U_n$

5-2- Directional earth fault protection :

- Setting current range : $(0.01 : 0.6) I_n$ with steps $\leq 0.01 I_n$
- Start voltage: $(2\%) U_n$
- Operate time in DT mode : $(0.05 : 3)$ sec with steps ≤ 0.01 sec
- Basic angle ϕ ($0 : 359^\circ$) at step $\leq 5^\circ$ or $(-90^\circ : +90^\circ)$ with Operation direction (Forward & Reverse)

The polarizing quantity shall be : the zero sequence voltage which the relay itself can reconstitute from the output voltage of the three single phase voltage transformers.

5-3- Three-phase non-directional over current protection :

5-3-1- Current stages :



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Low set over current $I > (0.05 - 5) I_n$ with steps $\leq 0.05 I_n$.
- High set over current $I \gg (0.1 - 25) I_n$ with steps $\leq 0.05 I_n$.

5-3-2 Operating time ranges :

- Definite time
- Low set over current stage $I > : (0.05 - 3) \text{ sec}$ with step $\leq 0.01 \text{ sec}$
- High set over current stage $I \gg : \text{instantaneous or operating time } (0 - 3) \text{ sec}$ with step 0.01 sec .

5-4- Non-directional earth fault protection :

5-4-1- Current stages :

- Low set earth fault $I_o > (0.01 - 0.6) I_n$ with steps $\leq 0.01 I_n$.
- High set earth fault $I_o \gg (0.01 - 2) I_n$ with steps $\leq 0.01 I_n$.

5-4-2-Operating time ranges :

- Definite time
- Low set earth fault $(0.05 - 3) \text{ sec}$ with steps $\leq 0.01 \text{ sec}$.
- High set earth fault $(0 - 3) \text{ sec}$ with steps $\leq 0.05 \text{ sec}$.

5-5- Voltage Protection :(optional)

5-5-1- Over voltage stages $U > :$

- Over voltage setting range : $60 : 150 \% U_n$ (with steps $\leq 2.5 \% U_n$)
- Definite time characteristics :
Operating time : $0.05:300$ (step 0.02 sec.) (with steps $\leq 0.05 \text{ sec.}$)



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

5-5-2- Under voltage stages $U < :$

- Under voltage setting range : 30 : 120 % U_n (with steps $\leq 2.5\% U_n$)
- Definite time characteristics :
Operating time : 0.05 : 300 (step 0.02) (with steps ≤ 0.05 sec.)

6- Diagnosis and Supervision :

- 6-1- Supervision function for energizing current input circuit.
- 6-2- Supervision function for energizing voltage input circuit.
- 6-3- Trip circuit supervision.
- 6-4- Self test diagnostic

7-Measurement :

- 7-1- Current measurement ($I_R-I_S-I_T-I_N$)
- 7-2- Voltage measurement ($V_R-V_S-V_T$) & ($V_{rs} - V_{st} - V_{tr}$)

8-Communications :

The relay shall support local and remote communication

- Local communication with P.C or laptop (RS 232)
- Remote communication through RS485 or fiber optic.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

II-outgoing & bus coupler Cubicles:

3 - Phase Non-directional Over current + Earth fault & Short circuit relay

The required is a combined non directional (over current + earth fault & short circuit) protection relay and timer in one unit connected to three current transformer of the protected object and suitable for resistance earthed neutral system, in solidly grounded system the residual over current protection can be used on feeder of any length and fulfilling the following requirements and specifications:

1- GENERAL CHARACTERISTICS:-

- 1.1-The protection relay is based on a multiprocessor environment.
- 1.2-The modern technology shall be applied in both hardware and software solution.
- 1.3-The relay must be designed for protection against (O.C. & E.F & S.C & Thermal) includes the following:-
 - Reset factors $\geq 95\%$
 - Operation temp. From -5°C to $+55^{\circ}\text{C}$.
 - Humidity $> 90\%$ non-condensation
 - Auxiliary supply : supply : $110\text{ V D.C} \pm 20\%$
 - It shall be able to operate without intended delay.
 - Definite and inverse time characteristics according to IEC standards.
 - Four programmable LEDs (besides LEDs of ready of protection and trip indication)
 - Self-monitoring and blocking for internal faults with alarm contact and led indication.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Tripping circuit failure supervision.
- Operation indication, with hand reset and automatic reset for tripping order and include one LED for each fault type and monitoring
- The three phases over current unit and the earth fault measure the phase current and the neutral current of the protected object.
- The relay shall be flush mounted .
- The relay must be digital type & compatible with any SCADA system (option).
- Nominal current $I_n = 5A/1A$ frequency = 50 HZ.
- Digital inputs :
 - At least 2 binary input.
 - Binary input voltage $110\text{ VDC} \pm 20\%$
- Output contacts (Tripping contacts) .
 - Rated voltage $110\text{ VDC} \pm 20\%$ with ripple content (3-4) %.
 - Continuous carrying capacity 5 A DC.
 - Making and carrying current capacity for 0.5 sec $\geq 25\text{ A}$.
 - Make and carry for 3.0 sec $\geq 10\text{ A}$
 - Free contacts at least (1 N.O for O.C & 1 N.O for E.F and 1 N.C for signaling).
 - Two trip contacts and two configurable per relay
- Phase discontinuity protection for three phases
- Three-phase thermal protection .



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- The relay must have the inrush current blocking based on the second order harmonic percentage which can be used in all protection stages , without using multiplayer
- All parameters and events can be set or read by P.C and manually by key pad
- Smaller dimensions preferred
- Tender must supply with offer detailed instruction and catalogue
- Conductor broken from consumer (open circuit fault) : 46 bc (As option).
- Zero sequence ground (As options)
- The relay should be compatible with SCADA system (option)
- The relay must include event recorder and shall be read by screen.
- Original software, firmware, plug connections to (PC and / or laptop) for all protection relays and test plugs should provide by the contractor.
- Continuous self-supervision and diagnosis of electronics and software
- User-selectable password protection for HMI
- Display of primary/secondary (current & voltage) values
- Housing degree of protection IP 52 according to IEC 60529
- The relay should be supported by standard protocol IEC 60870-5-103 **if needed**
- The relay should be supplied with its software and cables needed for communicate it with any computers.
- The output contact functions freely configurable for desired operation.
- The relay shall have two setting tables.
- High impedance restricted earth fault(As options).
- Circuit breaker failure protection (0.01.....1) sec.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

1. 4 The relay must be include the following protection:

- $I >$ Three-phase low-set over current stage with definite-time and inverse definite minimum time characteristic (IDMT) (as option)
- $I \gg$ Three-phase high-set over current stage with the instantaneous or definite time characteristic.
- $I_o >$ low-set non-directional earth –fault stage with definite-time or inverse definite minimum time characteristic (IDMT).
- $I_o \gg$ High-set non-directional earth-fault stage with instantaneous or definite – time characteristic.

2- Non-volatile memory for the following:

The protection relay should be equipped with a nonvolatile memory for preserving important data during auxiliary supply breaks. The memory does not need batteries, and a lifelong service is guaranteed and Following data is stored:

- Date and Time.
- Setting values
- Disturbance Records
- Recorded last five fault data.
- Last 50 events

Recorded values are stored from start, trip and other important events.

The last 50 events that occurred before the auxiliary power supply breaks can be viewed in the HMI event buffer. Time and date for the events are also restored.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Disturbance recorder with the following features:

- Nonvolatile memory for disturbance record all data shall be kept even for switch off D.C. power supply.
- Four analogue channels for (Ia , Ib , Ic , Io) – (Va , Vb , Vc , Vo) (voltages is option) .
- Eight digital channels at least .
- The relay shall keep minimum of 5 records which can be stored.
- Recording time capacity ≥ 15 sec.
- Recording time up to three seconds duration for each storage fault
Pre event and post event recording and memory time adjustable.

- Oscillography recorder : voltage and current waves must be recorded so that can be analyzed oscillography through PC for all abnormal .

- Fault recording from pre fault up to fault clearing stage with all electrical parameters.

3- The real-time clock

The relay must be supported by a real-time clock (RTC) which used for time stamping of events. It is also running during auxiliary power breaks.

When the supply is RE-established, the relay sets the right time and new events are stamped accordingly.

4- Setting values and ranges:-

4-1- Current stages (over current)



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Low set over current $I > (0.05 - 5) I_n$ with steps $\leq 0.05 I_n$.
- High set over current $I >> (0.1 - 25) I_n$ with steps $\leq 0.05 I_n$

4-2- Operating time ranges (over current)

- Operate time in DT mode: $(0.05 - 3) \text{ sec}$ with step $\leq 0.01 \text{ sec}$
- Low set over current stage $I > : 0.05 - 3 \text{ sec}$ with step $\leq 0.01 \text{ sec}$
- High set over current stage $I >> : \text{instantaneous or operating time } (0-3) \text{ sec.}$

4-3- Current stages (earth fault)

- Low set earth fault $I_o > : (0.01 - 0.6) I_n$ with steps $\leq 0.01 I_n$.
- High set earth fault $I_o >> : (0.1 - 10) I_n$ with steps $\leq 0.01 I_n$.
- Sensitive earth fault : $(0.005 - 0.5) I_n$ (as option)

4-4- Operating time ranges(earth fault)

- Definite time
- Low set earth fault $I_o > : (0.05 - 3) \text{ sec}$ with steps $\leq 0.05 \text{ sec}$.
- High set earth fault $I_o >> : (0 - 3) \text{ sec}$ with steps $\leq 0.05 \text{ sec}$.
- Sensitive earth fault : $(0 - 3) \text{ sec}$ (as option)

5- Standard tests

A- Insulation tests

- Dielectric tests : according to the (IEC 60255/5 & 60225/5 or equivalent) tests
voltage 2 KV 50 HZ, 60 sec



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- Impulse voltage test according to the (IEC 60255/5 & 60225/5 or equivalent) Tests voltage 5KV, unipolar impulses, waveform 1.2/50 μ s
- Insulation resistance measurements according to the IEC60255/5.

B- High frequency interference withstands:

According to class III IEC 60225/5.

C- Over load

- Current circuit temporary > 60 In for 1.0 sec
- Polarization temporary > 1.3 Ur max (Ur = 3 Uo)

D- Mechanical tests

- Vibration test (sinusoidal) according to the IEC 62271-103.
- Shock and bump test according to the IEC 62271-103.

6- diagnostis and Supervision

6-1- Supervision function for energizing current input circuit.

6-2- Trip circuit supervision.

6-3- Self test diagnostic.

7- Measurement

7-1- Display primary current values (IR-IS-IT-IN).

7-2- Voltage measurement (V_R - V_S - V_T) & ($V_{rs} - V_{st} - V_{tr}$).



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

7-3-Transient disturbance records.

8-Communications :

The relay shall support local and remote communication

- Local communication with P.C or laptop (RS 232)

Remote communication through RS485 or fiber optic.

Standards

The offered equipment is designed , manufactured and tested in accordance with the applicable

IEC standards:

- | | |
|------------------------------|---------------|
| - C.B's | IEC 62271-100 |
| - Switch board | IEC 62271-200 |
| - Contactors | IEC 62271-106 |
| - Current transformer | IEC 61869-1,2 |
| - Voltage transformer | IEC 61869-1,3 |
| - Relays | IEC 60255 |
| - Insulation coordination | IEC 60071 |
| -Vibration test (sinusoidal) | IEC 62271-103 |



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

Minimum requirements of spare Parts: (option)

- 1- One incoming circuit breaker complete with truck.
- 2- One out going circuit breaker complete with truck.
- 3- Three current transformer 200-400/5-5 A(as delivered).
- 4- Three current transformer 300-600/5-5 A(as delivered).
- 5- Three voltage transformer (as delivered).
- 6- One 3 pole earthing switch (as delivered).
- 7- (9) Digital Ammeter (as delivered).
- 8- One Digital voltmeter (as delivered).
- 9- One combined directional & non direction (O.C+E.F+S.C) relay as delivered for incoming cells.
- 10- One combined non direction (O.C+E.F+S.C) relay as delivered for outgoing cells.
- 11- (6) bus bar supporting insulators.
- 12- (6) H.V fuses for P.T (as delivered).
- 13- Two spring charging motor 110V.DC (as delivered) or control module in case of magnetic actuator.
- 14- Two space heater (as delivered).
- 15- Two tripping coil (as delivered) in case of spring charge motor C.B. type.
- 16- Two closing coil (as delivered) in case of spring charge motor C.B. type
- 17- Two Blocking coils for incoming cell (as delivered) in case of spring charge motor C.B. type
- 18- Two complete magnetic actuator units with three magnetic coils in case of magnetic actuator C.B. type (as delivered).

❖ Notice : the items from 3 to 18 should be delivered in suitable wooden container.

❖ شروط هامة يجب مراعاتها في جميع خلايا الموزع .

١. يجب ان لا يتغير وضع تشغيل قواطع الجهد المتوسط بخلايا الموزع(الفصل-التوصيل) في الحالات التالية:

- فصل الجهد المستمر عن جميع خلايا الموزع.
- فصل الجهد المستمر بأي خلية بالموزع.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

- فصل الجهد المستمر عن جهاز الوقاية بأي خلية.
- عند إعادة توصيل الجهد المستمر للموزع.
- عند إعادة توصيل الجهد المستمر لأي خلية.
- عند إعادة توصيل الجهد المستمر لجهاز الوقاية بأي خلية.
- ٢. يجب أن يعمل الموزع يدويا وكهربيا وليس أوتوماتيكيا مع مراعاة وجود الانترلوك بين خلايا الدخول وخلية الربط .
- ٣. يجب أن يتم تشغيل (توصيل وفصل) جميع القواطع (الخروج- الدخول – الربط بالموزع) يدويا وباب الخلية مغلق.
- ٤. يلزم عند فصل أحد مفتاحي الدخول التي تعمل بالتوازي بالموزع على قصر أن لا يتأثر المفتاح الآخر ويكون تشغيل كل مفتاح دخول مستقل عن الآخر وذلك للحفاظ على إستمرارية التغذية عن طريق أحد خطي التوازي .
- ٥. يلزم توريد قطع الغيار الخاصة بكل موزع على حده في صندوق خشبي (البند من ٣ – ١٨ قطع الغيار) مدون عليه رقم الموزع.
- ٦. يلزم أن يتم توريد مشتملات تركيب كل موزع على حدة في صندوق خشبي مدون عليه رقم الموزع.
- ٧. يجب أن تكون أجهزة الوقاية المستخدمة بالموزع قد سبق توريدها واستخدامها بشركات توزيع الكهرباء وأداؤها سليم بالموقع أما بالنسبة للأجهزة التي لم يسبق توريدها فيتم إختبارها من قطاع الوقاية المركزية وإعتمادها من الشركة القابضة للكهرباء مصر.
- ٨. يجب أن تكون أجهزة الوقاية لا تتأثر الذاكرة بها بفصل الجهد المساعد للتشغيل (DC volt).
- ٩. يجب تقديم عدد (٢) كتالوج أصلى للقواطع وأجهزة الوقاية والقياس ومحولات التيار والجهد والرسومات التنفيذية ودوائر الوقاية والقياس - عند التوريد.
- ١٠. ان تكون لمبات البيان ومفاتيح الفصل والتوصيل من النوع الذى يتحمل الخدمة الشاقة(من النوع LED).
- ١١. أن تكون تغذية جميع المهمات علي واجهة أي خلية مباشرة من الروزات داخل الخلية وبأسلاك كنترول مرنة بحيث تكون روزيته لكل سلك وعدم تركيب كباري (Bridges) بين المهمات (لمبات البيان المختلفة - مفاتيح التشغيل) علي باب الخلية لتجنب ظاهرة غياب الجهد المستمر عن هذه المهمات.
- ١٢. تركيب مفتاح عمومي ثلاثي (M.C.B) للسخان داخل خلية الربط وتقسيم الأحمال على كل وجه (Phase) مع تركيب مفتاح لكل سخان داخل كل خلية لحماية السخان.
- ١٣. جميع الروزات تصنع من مواد Halogen free غير قابلة للاشتعال .



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

١٤. يجب إن يتم تركيب ترامل مكبس لكل أسلاك الكنترول مع الترقيم.
١٥. يجب أن تكون الروزئات علي الدوائر الثانوية لمحول تيار مكونة من مجموعتين الأولي للقياس والثانية للوقاية وتكون مزودة بوسيلة لقفلهما في اي وقت لتغيير العدادات أو جهاز الوقاية.
١٦. يلزم أن تكون ألوان أسلاك محولات التيار مختلفة عن ألوان أسلاك محولات الجهد ومختلفة عن ألوان أسلاك دوائر الجهد المستمر DC .
١٧. ترقيم الروزئات بطريقة غير قابلة للمحو.
١٨. جميع الروزئات والترامل تصنع من نحاس عالي الجودة ويكون مطليا بالقصدير.
١٩. لكل موزع يجب توريد عدد ١ برنامج (software) نسخة تشغيل أصلية لأجهزة الوقاية المركبة بالموزع وعدد ١ حاسب شخصي متنقل حديث وكذا عدد ٤ كابل ربط بالحاسب الشخصي لكل موزع على أن يتم تدريب مهندسي الوقاية على استخدام برامج (software).
٢٠. أن يتم توصيل الملامسات المساعدة (Free contacts)(2NO+2NC) داخل القاطع وبأجهزة الوقاية الي روزئات خاصة بذلك داخل كل خلية مع ترقيمها.
٢١. (في حالة توريد موزعات (cassette type) يتم توريد عدد ٤ ترولي لكل موزع لإخراج وإدخال القواطع أثناء عملية الصيانة والمناورات.
٢٢. يجب أن يكون التصميم الميكانيكي والكهربائي لملفات الفصل والتوصيل بكل قاطع بالموزع محمي ذاتيا من الأتربة ولا يتأثر بها ولا يؤدي إلي زيادة زمن الفصل والتوصيل المقنن لأي قاطع.
٢٣. يفضل أن يكون غطاء ظهر الخلية من الخلف مرقم ومكون من أكثر من جزء لسهولة التعامل مع المكونات الداخلية للخلية أثناء عملية الصيانة أو تغييرها أو إحلال بعض منها.
٢٤. يلزم مراجعة وضبط نهاية مشوار الدخول للمفاتيح مع النقاط الثابتة علي البارة علي أن يتم تركيب Shunt release آخر لضمان استكمال مشوار الفصل في حالة الحاجة لذلك .
٢٥. يجب أن لا يفتح باب الخلية والمفتاح في وضع التشغيل.
٢٦. يجب أن يكون تصميم الموزع يحقق سهولة التعامل مع محولات التيار لاختبارها وكذلك محولات الجهد لسهولة تغيير فيوزاتها .
٢٧. يجب تاريض أبواب الخلايا وأجهزة القياس والوقاية .



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

٢٨. يجب أن يتوافر بكل خلية بالموزع (دخول – خروج – ربط) روزنات خالية لاستعمالها للربط علي التحكم الآلي من كل البنود التالية:

- الجهد المساعد المستمر
- نقاط أطراف الفصل والتوصيل
- الجهد المتغير A.C لمحاولات الجهد
- حالة القواطع
- دلالات الفصل بأجهزة الوقاية.

٢٩. يجب عمل عدد ٣ فتحات بجميع الخلايا لإمرار الخرطوم الحساس الخاص بالإطفاء التلقائي بغرف الباسبار – كاباتلات الكنترول بحيث يكون ارتفاع مستوى الخرطوم أعلى الباسبار وأعلى محولات التيار ١٠:١٢ سم.

٣٠. يجب ان تقوم سكينه الارضى بجميع خلايا الخروج والدخول بالموزع بتأريض رؤوس الكاباتلات بكل خلية اى تأريض محولات التيار من جهة رؤوس الكاباتلات وليس جهة الباسبار.

٣١. يجب أن تتم تجربة تشغيل خلايا الدخول والربط واختبار الانترلوك بينهما بدون وجود الجهد المتوسط على خلايا الدخول وذلك فى وضع الإختبار وفى وضع التشغيل.

٣٢. يتم توريد عدد ١ وصلة كاملة (socket + plug) منفصلة للقاطع للاختبارات لكل موزع وبدون سعر.

٣٣. يجب أن يتم تجهيز الخلايا بالآتي (As option) :

أ- أخذ وضع سكينه الأرضي وتوصيلها بجهاز الوقاية عن طريق (Binary Input).

ب- أخذ وضع المفتاح (On & Off) وتوصيلة لجهاز الوقاية عن طريق (Binary Input).

ت- أخذ وضع المفتاح (داخل - خارج) الخلية وتوصيلة لجهاز الوقاية عن طريق (Binary Input).

٣٤. يتم تزويد عدد ٢٥ % (spare) من كل نوع من الروزنات المركبة بكل خلية مع تزويد ٢٥ % (space) لإمكانية

التزويد مستقبلا عند الحاجة.(As option)



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 01-200 - 1

Date: 03-05-2016

APPENDIX

EQUIPMENT DATA

GUARANTEE TABLE NO.(1)

1- CUBICLES :-

- a- Maker's name
- b- Type
- c- Material
- d- Thickness of sheet steel mm.
- e- Dimensions:
- 1- Width mm.
- 2- Depthmm.
- 3- Heightmm.

2- BUS BARS :

- a- Material
- b- Number and Size per phase/.....mm x mm
- c- Minimum clearance between phases mm
- d- Minimum clearance between phase to earth mm
- e- Space between bus bar supports mm
- f- Maximum allowable current density at worst site conditionsA/mm²



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Date: 03-05-2016

g- Dynamic current KA.

h- Thermal current for 3 seconds KA.

i- Withstand impulse voltage (peak) KV.

j- One minute A.C 50 c/s test voltage KV.

3- CONNECTORS :-

a- Material

b- No. / size per phase/.....mm x mm

c- Minimum clearance between phases mm

d- Minimum clearance between phases to earth mm

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Signature:

Date:



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GUARANTEE TABLE NO (2)

CIRCUIT BREAKERS. 24.K.V

1. Maker 's name
2. Country of manufacture
3. Type
4. Rated breaking capacity at normal operating voltage (for 3 sec)KA
5. PARTICULARS :
 - Standard specification
 - Type of main contacts:
 - a-Fixed
 - b- Moving
 - Material of main contacts
 - Total travel of moving main contactmm
 - Type of arc control device
 - Material of arcing contacts
 - Material of arc control device
 - Current density in contact area at nominal currentA/mm²
 - Voltage drop across contacts at nominal currentm.V
 - Type of operating mechanism
 - a- Closing
 - b- Tripping



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-
- D.C voltage for closing and trippingV
- a- NominalV
- b- Minimum closingV
- c- Minimum trippingV
- d- Minimum clearance in air:
- ❖ Phases to earthmm
- ❖ Between phasesmm
- Creepage distance of insulationcm
- Weight of breaker completekg
- 6. PERFORMANCE:**
- Rated voltage between phasesKV
- Rated currentA
- Rated breaking capacity
- ❖ SymmetricalKA
- ❖ AsymmetricalKA
- Rated operating sequence
- Rated making capacity (peak)KA
- Short time rating:
- ❖ One second (for earthing switch)K.A.
- ❖ Three secondsK.A.



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EDMS 01-200 - 1

Date: 03-05-2016

- Total travel of moving main contactsmm
- Speed of breaker contactscm/sec.
- Time from energizing trip coil to contact SeparationCycles.
- Time for complete Separation of contactsCycles.
- Total break timeCycles.
- Total make timeCycles.

- Power required (at nominal Current or voltage)by:
 - ❖ Tripping coilWatts.
 - ❖ Closing coilWatts.

- Flash over voltage of bushing (dry)K.V

- One-minute power frequency:
 - ❖ Withstand voltage rms K.V

- 1.2/50 micro second impulse
 - ❖ Withstand voltage rms K.V

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Date:



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GUARANTEE TABLE NO (3) .

CURRENT TRANSFORMERS

- 1- Maker's name
- 2- Country of manufacture
- 3- Type
- 4- Standard specification
- 5- Service voltage at siteKV.
- 6- Highest system voltageKV.
- 7- Rated frequencyHZ
- 8- Rated ratio
- 9- Rated burden for:
 - a-..MeasuringVA
 - b-..RelayingVA
- 10- Accuracy class for (to write down error at the rated short circuit current factor).
 - a- Measuring
 - b- Relaying
- 11- Short circuit strength.
 - a- For one second



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EDMS 01-200 - 1

Date: 03-05-2016

b- For three secondsKA

.....KA

.....KA

12- Dynamic current (peak)

13- Over current factor

.....

14- Power frequency withstand voltage r.m.s

.....KV

15- Impulse test voltage 1.2/50,μs rms

.....KV

16- Approximate weight

.....Kg

17- Leakage path

.....Cm

We guarantee the data given above for the equipment offered.

Signature:

Date:



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GUARANTEE TABLE NO (4) .

POTENTIAL TRANSFORMERS

- | | |
|---|------------|
| 1- Maker's name | |
| 2- Country of manufacture | |
| 3- Type | |
| 4- Standard | |
| 5- Service voltage at site |K. V. |
| 6- Highest system voltage |K. V. |
| 7- Rated frequency |HZ |
| 8- Rated ratio | |
| 9- Rated burden |VA |
| 10- Accuracy class for: | |
| a- First core | |
| b- Second core | |
| 11- rated voltage factor and corresponding rated time | |
| 12- Power frequency withstand voltage(r.m.s) |K. V. |
| 13- Impulse test voltage (1.2/50) μ S (r.m.s) |K. V. |
| 14- Approximate weight | KGS |



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Date: 03-05-2016

15- Leakage Path

.....cm

We guarantee the data given above for the equipment offered.

Signature:

Date:



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GUARANTEE TABLE NO. (5)

PROTECTIVE RELAYS.

DESCRIPTION:

- 1- Maker 's name
- 2- Country of manufacture
- 3- Type and designation
- 4- Standard specification with which
relay complies
- 5- Relays characteristics
- 6- Range of relay setting I.....(A)T.....(Sec)
- 7- Current rating of relay coil A
- 8- Voltage rating of relay coilV
- 9- Range of Auxiliary supplyDC (V)

We guarantee the data given above for the equipment offered

Signature:

Date: